

Safety Data Sheet

ULTRABOND TURF PU 1K

Safety Data Sheet dated: 22/06/2022 - version 3

Date of first edition: 03/05/2017



Section 1: Identification

GHS Product identifier

Mixture identification:

Trade name: ULTRABOND TURF PU 1K

Trade code: 902333

Recommended use of the chemical and restrictions on use

Recommended use: Polyurethane-based adhesive

Uses advised against: no data available

Supplier's details

Company: MAPEI AUSTRALIA Pty Ltd

180 Viking Drive Wacol QLD 4076 Australia

T. +61 7 32765000 (Mon-Fri 8am to 4.30pm)

F. +61 7 32765076

Responsible: sales@mapei.com.au

Emergency phone number

Australian Poisons Information Centre 24 Hour Service 13 11 26

Police or Fire Brigade 000

Section 2: Hazard(s) identification



Classification of the Hazardous chemical

Flammable liquid, Category 4

Skin irritation, Category 2

Eye irritation, Category 2A

Respiratory Sensitisation, Category 1

Skin Sensitisation, Category 1B

Carcinogenicity, Category 2

Specific target organ toxicity — repeated exposure, Category 2

Combustible liquid

Causes skin irritation.

Causes serious eye irritation.

May cause allergy or asthma symptoms or breathing difficulties if inhaled.

May cause an allergic skin reaction.

Suspected of causing cancer.

May cause damage to organs through prolonged or repeated exposure.

Adverse physicochemical, human health and environmental effects:

No other hazards

GHS label elements, including precautionary statements

Pictograms and Signal Words



Danger

Hazard statements:

H227	Combustible liquid
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H319	Causes serious eye irritation.
H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H351	Suspected of causing cancer.
H373	May cause damage to organs through prolonged or repeated exposure.

Precautionary statements:

P201	Obtain special instructions before use.
P202	Do not handle until all safety precautions have been read and understood.
P210	Keep away from heat/sparks/open flames/hot surfaces. — No smoking.

P260	Do not breathe mist/vapours/spray.
P264	Wash hands thoroughly after handling.
P280	Wear protective gloves/clothing and eye/face protection.
P281	Use personal protective equipment as required.
P285	In case of inadequate ventilation wear respiratory protection.
P302+P352	IF ON SKIN: Wash with plenty of soap and water.
P304+P341	IF INHALED: If breathing is difficult, remove victim to fresh air and keep at rest in a position comfortable for breathing.
P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P308+P313	IF exposed or concerned: Get medical advice/attention.
P314	Get medical advice/attention if you feel unwell.
P321	Specific treatment (see supplementary instructions on this label)
P332+P313	If skin irritation occurs: Get medical advice/attention.
P333+P313	If skin irritation or rash occurs: Get medical advice/attention.
P337+P313	If eye irritation persists: Get medical advice/attention.
P342+P311	If experiencing respiratory symptoms: Call a POISON CENTER or doctor/physician.
P362	Take off contaminated clothing and wash before reuse.
P370+P378	In case of fire, use a dry powder fire extinguisher for extinction.
P403+P235	Store in a well-ventilated place. Keep cool.
P501	Dispose of contents/container in accordance with applicable regulations.

Other hazards which do not result in a classification

Other Hazards: No other hazards

Section 3: Composition and information on ingredients

Substances

no data available

Mixtures

Mixture identification: ULTRABOND TURF PU 1K

Hazardous components within the meaning of the "Australian Work Health and Safety (WHS)" regulation and related classification:

Concentration (%) w/w)	Name	Ident. Numb.	Classification	Registration Number
≥25 - <50 %	Calcium carbonate	CAS:471-34-1 EC:207-439-9		Exempted
≥10 - <20 %	Reaction mass of 4,4'-methylenediphenyl diisocyanate and o-(p-isocyanatobenzyl)phenyl isocyanate / methylene diphenyl diisocyanate	CAS:26447-40-5 EC:247-714-0 Index:615-005-00-9	Acute Tox. 4, H332; Skin Irrit. 2, H315; Eye Irrit. 2A, H319; Resp. Sens. 1, H334; Skin Sens. 1B, H317; Carc. 2, H351; STOT SE 3, H335; STOT RE 2, H373	01-2119457015-45-XXXX
≥5 - <10 %	2-(2-butoxyethoxy)ethyl acetate	CAS:124-17-4 EC:204-685-9	Aquatic Acute 3, H402	01-2119475110-51-XXXX

Section 4: First-aid measures

Description of necessary first-aid measures

In case of skin contact:

- Immediately take off all contaminated clothing.
- Remove contaminated clothing immediately and dispose of safely.
- After contact with skin, wash immediately with soap and plenty of water.

In case of eyes contact:

- After contact with the eyes, rinse with water with the eyelids open for a sufficient length of time, then consult an ophthalmologist immediately.
- Protect uninjured eye.

In case of Ingestion:

- Do not induce vomiting, get medical attention showing the SDS and the hazard label.

In case of Inhalation:

- Remove casualty to fresh air and keep warm and at rest.

Symptoms caused by exposure

Eye irritation
Eye damages
Skin Irritation
Erythema

Medical attention and special treatment

In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible).

Treatment:

(see paragraph 4.1)

Section 5: Firefighting measures

Suitable extinguishing media

None in particular.

Specific hazards arising from the chemical

Do not inhale explosion and combustion gases.
Burning produces heavy smoke.
Hazardous combustion products: no data available
Explosive properties: no data available
Oxidizing properties: no data available

Special protective equipment and precautions for fire-fighters

Use suitable breathing apparatus.
Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Move undamaged containers from immediate hazard area if it can be done safely.

Section 6: Accidental release measures

Personal precautions, protective equipment and emergency procedures

Wear personal protection equipment.
Remove persons to safety.

Environmental precautions

Do not allow to enter into soil/subsoil. Do not allow to enter into surface water or drains.
Limit leakages with earth or sand.

Methods and materials for containment and cleaning up

Suitable material for taking up: absorbing material, organic, sand
Retain contaminated washing water and dispose it.

Section 7: Handling and storage

Precautions for safe handling

Avoid contact with skin and eyes, inhalation of vapours and mists.
Exercise the greatest care when handling or opening the container.
Don't use empty container before they have been cleaned.
Before making transfer operations, assure that there aren't any incompatible material residuals in the containers.
Contaminated clothing should be changed before entering eating areas.
Do not eat or drink while working.
See also section 8 for recommended protective equipment.

Conditions for safe storage, including any incompatibilities

Always keep in a well ventilated place.
Store at below 20 °C. Keep away from unguarded flame and heat sources. Avoid direct exposure to sunlight.
Keep away from unguarded flame, sparks, and heat sources. Avoid direct exposure to sunlight.
Keep away from food, drink and feed.

Incompatible materials:

None in particular.

Instructions as regards storage premises:

Cool and adequately ventilated.

Section 8: Exposure controls and personal protection

Control parameters – exposure standards, biological monitoring

List of components with OEL value

OEL Type	Country	Ceiling	Long Term mg/m3	Long Term ppm	Short Term mg/m3	Short Term ppm	Note
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Calcium carbonate CAS: 471-34-1	AUS	AUSTRALIA	10			
	National	FRANCE	10			
	National	PORTUGAL	10			
	National	LATVIA	6			
Reaction mass of 4,4'-methylenediphenyl diisocyanate and o-(p-isocyanatobenzyl)phenyl isocyanate / methylene diphenyl diisocyanate CAS: 26447-40-5	OSHA	C		0.2	0.02	
	National	GREECE	0.2	0.02	0.2	0.02
	National	BULGARIA	0.05		0.07	
	National	LITHUANIA	0.05	0.005		
	National	LITHUANIA	C		0.1	0.01
2-(2-butoxyethoxy)ethyl acetate CAS: 124-17-4	National	SWEDEN	130	15	250	30
	National	SWITZERLAND	85	10	127.5	15
	National	GERMANY	67	10		
	National	LITHUANIA	130	15	250	30
	National	SLOVENIA	67.5	10	101.2	15

Predicted No Effect Concentration (PNEC) values

	PNEC Limit	Exposure Route	Exposure Frequency	Remark
Calcium carbonate CAS: 471-34-1	100 mg/l	Microorganisms in sewage treatments		
2-(2-butoxyethoxy)ethyl acetate CAS: 124-17-4	0.108 mg/l	Fresh Water		
	0.011 mg/l	Marine water		
	0.8 mg/kg	Freshwater sediments		
	0.08 mg/kg	Marine water sediments		
	0.29 mg/kg	Soil		
	70 mg/kg	Oral		
	100 mg/l	Microorganisms in sewage treatments		

Derived No Effect Level. (DNEL)

	Worker Industrial	Worker Professional	Consumer	Exposure Route	Exposure Frequency	Remark
Calcium carbonate CAS: 471-34-1	6.36 mg/m3	1.06 mg/m3		Human Inhalation	Long Term, local effects	
		6.1 mg/kg		Human Oral	Long Term, systemic effects	
		6.1 mg/kg		Human Oral	Short Term, systemic effects	
2-(2-butoxyethoxy)ethyl acetate CAS: 124-17-4	100 mg/kg	60 mg/kg		Human Dermal	Long Term (repeated)	
		7.9 mg/kg		Human Oral	Long Term (repeated)	
	85 mg/m3			Human Inhalation	Long Term, systemic effects	

Appropriate engineering controls

Appropriate engineering controls

no data available

Individual protection measures, such as personal protective equipment (PPE)

Eye protection:

Use close fitting safety goggles, don't use eye lens.

Protection for skin:

Use clothing that provides comprehensive protection to the skin, e.g. cotton, rubber, PVC or viton.

Protection for hands:

Suitable materials for safety gloves; AS/NZS 2161.10:

Polychloroprene - CR: thickness $\geq 0,5\text{mm}$; breakthrough time $\geq 480\text{min}$.

Nitrile rubber - NBR: thickness $\geq 0,35\text{mm}$; breakthrough time $\geq 480\text{min}$.

Butyl rubber - IIR: thickness $\geq 0,5\text{mm}$; breakthrough time $\geq 480\text{min}$.

Fluorinated rubber - FKM: thickness $\geq 0,4\text{mm}$; breakthrough time $\geq 480\text{min}$.

Use protective gloves that provides comprehensive protection, e.g. P.V.C., neoprene or rubber.

Respiratory protection:

Respiratory protection must be used where exposure levels exceed workplace exposure limits. Refer to AS/NZS 1715-1716 for information on selection and use of appropriate respiratory protection equipment.

Use respiratory protection where ventilation is insufficient or exposure is prolonged.

Use adequate protective respiratory equipment.

Section 9: Physical and chemical properties

Physical state: Liquid

Color: Green

Physical state Liquid

Color

Odour:

Green

Characteristic

pH: no data available

Melting point / freezing point: no data available

Initial boiling point and boiling range: no data available

Flash point: 62 °C (144 °F)

Evaporation rate: no data available

Flammability (Solid, Gas): no data available

no data available

Vapour pressure: no data available

Lower and upper explosion limit/flammability limits:

Vapour density: no data available

Relative density: 1.34 g/cm³

Solubility in water: Insoluble

Solubility in oil: partly soluble

Partition coefficient (n-octanol/water): no data available

Auto-ignition temperature: no data available

Decomposition temperature: no data available

Particle size: no data available

Kinematic viscosity: no data available

Particle size distribution: no data available

Particle characteristics:

Shape and aspect ratio: no data available

Specific surface area: no data available

VOC % (Volatile Organic Compound) : No data available

Section 10: Stability and reactivity

Reactivity

Stable under normal conditions

Chemical stability

no data available

Possibility of hazardous reactions

None.

Conditions to avoid

Stable under normal conditions.

Incompatible materials

None in particular.

Hazardous decomposition products

Section 11: Toxicological information

Information on toxicological effects

Toxicological information of the mixture:

a) acute toxicity	Not classified Based on available data, the classification criteria are not met
b) skin corrosion/irritation	The product is classified: Skin irritation, Category 2(H315)
c) serious eye damage/irritation	The product is classified: Eye irritation, Category 2A(H319)
d) respiratory or skin sensitisation	The product is classified: Respiratory Sensitisation, Category 1(H334), Skin Sensitisation, Category 1B(H317)
e) germ cell mutagenicity	Not classified Based on available data, the classification criteria are not met
f) carcinogenicity	The product is classified: Carcinogenicity, Category 2(H351)
g) reproductive toxicity	Not classified Based on available data, the classification criteria are not met
h) STOT-single exposure	Not classified Based on available data, the classification criteria are not met
i) STOT-repeated exposure	The product is classified: Specific target organ toxicity — repeated exposure, Category 2(H373)
j) aspiration hazard	Not classified Based on available data, the classification criteria are not met

Toxicological information on main components of the mixture:

Calcium carbonate	a) acute toxicity	LD50 Oral Rat > 2000 mg/kg LC50 Inhalation Rat > 3 mg/l LD50 Skin Rat > 2000 mg/kg 4h LD50 Oral Rat = 6450 mg/kg	
	g) reproductive toxicity	NOAEL Rat = 1000 mg/kg	
Reaction mass of 4,4'-methylenediphenyl diisocyanate and o-(p-isocyanatobenzyl)phenyl isocyanate / methylene diphenyl diisocyanate	a) acute toxicity	LD50 Skin Rabbit > 10000 mg/kg LC50 Inhalation Rat = 490 mg/m3 4h LD50 Oral Rat > 10000 mg/kg	
2-(2-butoxyethoxy)ethyl acetate	a) acute toxicity	LD50 Oral Rat = 11920 mg/kg LD50 Skin Rabbit = 5400 mg/kg LD50 Skin Rabbit = 14500 mg/kg LC50 Inhalation Rat = 72500 mg/m3 4h LD50 Oral Rat = 6500 mg/kg	
	i) STOT-repeated exposure	NOAEL Oral Rat = 315 mg/kg	90 d
		NOAEL Skin Rat = 2400 mg/kg	13 w
		NOAEL Inhalation Rat = 118 mg/m3	90 d

Section 12: Ecological information

Ecotoxicity

Adopt good working practices, so that the product is not released into the environment.

Eco-Toxicological Information:

List of Eco-Toxicological properties of the product

Not classified for environmental hazards

Based on available data, the classification criteria are not met

List of components with eco-toxicological properties

Component	Ident. Numb.	Ecotox Infos
Calcium carbonate	CAS: 471-34-1 - EINECS: 207-439-9	c) Bacteria toxicity : NOEC Bacteria = 1000 mg/L 3 d) Terrestrial toxicity : LC50 > 1000 mg/kg d) Terrestrial toxicity : NOEC = 1000 mg/kg - 28 d e) Plant toxicity : NOEC = 1000 mg/kg - 21 d
Reaction mass of 4,4'-methylenediphenyl diisocyanate and o-(p-isocyanatobenzyl)phenyl isocyanate / methylene diphenyl diisocyanate	CAS: 26447-40-5 - EINECS: 247-714-0 - INDEX: 615-005-00-9	d) Terrestrial toxicity : LC50 Worm Eisenia foetida > 1000 mg/kg 14d IUCLID d) Terrestrial toxicity : NOEC Worm Eisenia foetida >= 1000 mg/kg 14d IUCLID
2-(2-butoxyethoxy)ethyl acetate	CAS: 124-17-4 - EINECS: 204-685-9	a) Aquatic acute toxicity : EC50 Algae = 1570 mg/L 72h a) Aquatic acute toxicity : LC50 Fish Brachydanio rerio 50 mg/L 96h IUCLID a) Aquatic acute toxicity : LC50 Daphnia magna = 665 mg/L 48h IUCLID

Persistence and degradability

no data available

Bioaccumulative potential

no data available

Mobility in soil

no data available

Other adverse effects

no data available

Section 13: Disposal considerations

Disposal methods

The generation of waste should be avoided or minimized wherever possible. Recover if possible.

Methods of disposal:

Disposal of this product, solutions, packaging and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements.

Dispose of surplus and nonrecyclable products via a licensed waste disposal contractor.

Do not dispose of waste into sewers.

Disposal considerations:

Do not allow to enter drains or watercourses.

Dispose of product according to all federal, state and local applicable regulations.

If this product is mixed with other wastes, the original waste product code may no longer apply and the appropriate code should be assigned.

Dispose of containers contaminated by the product in accordance with local or national legal provisions. For further information, contact your local waste authority.

Special precautions:

This material and its container must be disposed of in a safe way. Care should be taken when handling untreated empty containers.

Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

Empty containers or liners may retain some product residues. Do not re-use empty containers.

Section 14: Transport information

UN number

NA1993

UN proper shipping name

ADG-Shipping Name: N.A.

ADR-Shipping Name: -

IATA-Technical name: -

IMDG-Technical name: -

Transport hazard class(es)

ADG-Class: -

ADR-Class: -

IATA-Class: -

IMDG-Class: -

Packing group, if applicable

ADG-Packing Group: -

ADR-Packing Group: -

IATA-Packing group: -

IMDG-Packing group: -

Environmental hazards

ADG-Environmental Pollutant: No

Marine pollutant: No

no data available

Special precautions for user

IATA-Subsidiary hazards:

IMDG-Subsidiary hazards:

no data available

Additional Information

no data available

HazChem Code/Emergency Action code

no data available

Section 15: Regulatory information

Safety, health and environmental regulations specific for the product in question

This Safety Data Sheet has been prepared according to the Australian Work Health and Safety (WHS) act and the Code of Practice on preparation of safety data sheets for Hazardous Chemicals.

AICS: all components are listed

Section 16: Any other relevant information

Code	Description
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H315	Causes skin irritation.
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H317	May cause an allergic skin reaction.
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H319	Causes serious eye irritation.
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H332	Harmful if inhaled.
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H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
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H335	May cause respiratory irritation.
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H351	Suspected of causing cancer.
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H373	May cause damage to organs through prolonged or repeated exposure.
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H402	Harmful to aquatic life
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Code	Hazard class and hazard category	Description
3.1/4/Inhal	Acute Tox. 4	Acute toxicity (inhalation), Category 4
3.2/2	Skin Irrit. 2	Skin irritation, Category 2
3.3/2A	Eye Irrit. 2A	Eye irritation, Category 2A
3.4.1/1	Resp. Sens. 1	Respiratory Sensitisation, Category 1
3.4.2/1B	Skin Sens. 1B	Skin Sensitisation, Category 1B
3.6/2	Carc. 2	Carcinogenicity, Category 2
3.8/3	STOT SE 3	Specific target organ toxicity — single exposure, Category 3
3.9/2	STOT RE 2	Specific target organ toxicity — repeated exposure, Category 2

This document was prepared by a competent person who has received appropriate training.

Main bibliographic sources:

ECDIN - Environmental Chemicals Data and Information Network - Joint Research Centre, Commission of the European Communities

SAX's DANGEROUS PROPERTIES OF INDUSTRIAL MATERIALS - Eight Edition - Van Nostrand Reinold

The information contained herein is based on our state of knowledge at the above-specified date. It refers solely to the product indicated and constitutes no guarantee of particular quality.

It is the duty of the user to ensure that this information is appropriate and complete with respect to the specific use intended.

This SDS cancels and replaces any preceding release.

Legend to abbreviations and acronyms used in the safety data sheet:

ACGIH: American Conference of Governmental Industrial Hygienists

ADR: European Agreement concerning the International Carriage of Dangerous Goods by Road.

AND: European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways

ATE: Acute Toxicity Estimate

ATEmix: Acute toxicity Estimate (Mixtures)

BCF: Biological Concentration Factor

BEI: Biological Exposure Index

BOD: Biochemical Oxygen Demand

CAS: Chemical Abstracts Service (division of the American Chemical Society).

CAV: Poison Center

CE: European Community

CLP: Classification, Labeling, Packaging.

CMR: Carcinogenic, Mutagenic and Reprotoxic

COD: Chemical Oxygen Demand

COV: Volatile Organic Compound

CSA: Chemical Safety Assessment

CSR: Chemical Safety Report

DMEL: Derived Minimal Effect Level

DNEL: Derived No Effect Level.

DPD: Dangerous Preparations Directive

DSD: Dangerous Substances Directive

EC50: Half Maximal Effective Concentration

ECHA: European Chemicals Agency

EINECS: European Inventory of Existing Commercial Chemical Substances.

ES: Exposure Scenario

GefStoffVO: Ordinance on Hazardous Substances, Germany.

GHS: Globally Harmonized System of Classification and Labeling of Chemicals.

IARC: International Agency for Research on Cancer

IATA: International Air Transport Association.

IATA-DGR: Dangerous Goods Regulation by the "International Air Transport Association" (IATA).

IC50: half maximal inhibitory concentration

ICAO: International Civil Aviation Organization.

ICAO-TI: Technical Instructions by the "International Civil Aviation Organization" (ICAO).

IMDG: International Maritime Code for Dangerous Goods.

INCI: International Nomenclature of Cosmetic Ingredients.

IRCCS: Scientific Institute for Research, Hospitalization and Health Care

KAFH: KAFH

KSt: Explosion coefficient.

LC50: Lethal concentration, for 50 percent of test population.

LD50: Lethal dose, for 50 percent of test population.

LDLo: Leathal Dose Low

N.A.: Not Applicable

N/A: Not Applicable

N/D: Not defined/ Not available

NA: Not available

NIOSH: National Institute for Occupational Safety and Health

NOAEL: No Observed Adverse Effect Level

OSHA: Occupational Safety and Health Administration.

PBT: Persistent, Bioaccumulative and Toxic

PGK: Packaging Instruction

PNEC: Predicted No Effect Concentration.

PSG: Passengers

RID: Regulation Concerning the International Transport of Dangerous Goods by Rail.

STEL: Short Term Exposure limit.

STOT: Specific Target Organ Toxicity.

TLV: Threshold Limiting Value.

TWATLV: Threshold Limit Value for the Time Weighted Average 8 hour day. (ACGIH Standard).

vPvB: Very Persistent, Very Bioaccumulative.

WGK: German Water Hazard Class.

Paragraphs modified from the previous revision:

- Safety Data Sheet
- 1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY/UNDERTAKING
- 2. HAZARDS IDENTIFICATION
- 3. COMPOSITION/INFORMATION ON INGREDIENTS
- 4. FIRST AID MEASURES
- 5. FIRE-FIGHTING MEASURES
- 6. ACCIDENTAL RELEASE MEASURES
- 7. HANDLING AND STORAGE
- 8. EXPOSURE CONTROLS/PERSONAL PROTECTION
- 9. PHYSICAL AND CHEMICAL PROPERTIES
- 10. STABILITY AND REACTIVITY
- 11. TOXICOLOGICAL INFORMATION
- 12. ECOLOGICAL INFORMATION
- 14. TRANSPORT INFORMATION
- 16. OTHER INFORMATION